

# Comparative Analysis of Traffic Accident Detection with Emphasis on Explainability of DL Models

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**Abstract:** In this study, we conducted an analysis utilizing machine learning techniques to distinguish between images depicting traffic accidents and those that do not. We gathered a dataset comprising samples of both accident and non-accident scenarios, and proceeded to employ machine learning and deep learning models including Convolutional Neural Networks (CNNs), Convolutional Neural Networks (CNNs) with Long Short-Term Memory (LSTM), Artificial Neural Networks (ANNs), Random Forests, Decision Trees, Support Vector Machines (SVMs), and You Only Look Once version 8 (YOLO v8). Through extensive training of these models on the dataset, we aimed to develop classifiers capable of accurately discerning the occurrence of accidents in images. To assess the efficacy and interpretability of these models, we employed SHAP (SHapley Additive exPlanations), an Explainable AI framework, which offers greater clarity to the decision-making process of machine learning models. Our findings illuminate the performance and interpretability of the employed models, supplying valuable insights into their capability to accurately classify accident-related imagery.

**Index terms:** Convolutional Neural Networks (CNN);Long Short term Memory (LSTM);Artificial Neural Networks (ANN);YOLO;Random Forest; Decision Tree;Support Vector Machines (SVM);SHAP;

## I. INTRODUCTION

The rising number of traffic accidents around the world is becoming a significant concern for public safety. Accurate image classification is pivotal in numerous applications, including autonomous vehicles, surveillance systems, and accident prevention technologies.

Machine learning and Deep learning technologies have made meaningful improvements in image classification, and multiple models have undergone training to identify accident-related images. However, identifying the most effective models is essential in accurately distinguishing between accident-related and non-accident-related images.

An extensive analysis was carried out to evaluate the effectiveness of different machine learning and deep learning models in categorizing photos as accidents or non-accidents in order to fill this research gap. Convolutional neural networks (CNNs), CNNs with long short-term memory (LSTM), artificial neural networks (ANNs), Random forests, decision trees,

Support Vector Machines (SVMs), and YOLO v8 are among the models assessed in this work. We used Explainable AI techniques during the model training process, which enabled us to determine the rationale for the model's classification of an accident vs not.

## II. RELATED WORK

Highway accidents incur significant delays and costs, necessitating prompt detection for traffic management and risk reduction. In recent years, there has been a growing focus on intelligent systems for accident detection, exploring diverse methodologies and technologies to enhance detection system efficiency and accuracy, aiming to mitigate the impact of accidents on public safety and traffic flow. Researchers and scientists are presenting various methods to aid in the identification of accidents. These methods include deep learning models, mobile applications, VANET, GPS, and GSM technologies, as well as the use of smartphones to identify traffic accidents.

Arifeen et al. (2023)[3] investigated the application of Convolutional Neural Networks (CNNs) for accident detection in traffic videos. They evaluated multiple architectures, including GoogLeNet, AlexNet, and VGGNet, achieving high accuracy through the use of One-Class Support Vector Machines (OCSVMs) for anomaly detection.

A machine learning-based system that integrates vibrating and ultrasonic sensors with GPS and GSM modules was presented by B.K.M et al. in 2021 [2]. Their approach aimed to accurately identify accidents by analyzing data from nearby vehicles, focusing on recognizing abnormal behaviors.

In a different approach, Dogru & Subasi (2018)[4] focused on real-time accident detection within Intelligent Transportation Systems (ITS) by employing Random Forest classifiers and Vehicular Ad Hoc Networks (VANETs). Equipping probe vehicles with RF enabled proactive measures to prevent secondary accidents.

The study by Yadav et al. (2020)[9] proposes an accident detection system employing CNN with LSTM units, achieving accuracies up to 97.76%. This study makes progress in accident identification using Deep learning techniques, but challenges related to the vanishing gradient problem may impact accuracy.

Additionally, Ki et al. (2018)[5] presented a neural network-based approach that leveraged traffic data for incident detection, offering a cost-effective alternative to traditional surveillance systems.

These studies collectively demonstrate a diverse range of methodologies and technologies aimed at improving accident detection and management in the highway system.

### III. METHODOLOGY

This study utilized a systematic way to classify traffic accident images through the implementation of diverse classification algorithms. The methodology involved several essential steps, including data collection, model implementation, training, testing, evaluation using standard metrics, and interpretation of results utilizing an explainable AI framework.

#### 1. Data Collection

The initial phase centered on obtaining datasets containing both traffic accident and non-accident images. The images were sourced from the Roboflow Universe website, which offers a vast selection of open-source datasets for computer vision. This dataset helped to reduce bias and ensure the accuracy and reliability of the results.

#### 2. Model Implementation

Various ML and DL algorithms were implemented to classify the acquired images:

#### **Convolutional Neural Network (CNN) with Long Short-Term Memory (LSTM):**

CNNs excel in obtaining spatial features from images, while LSTMs are adept at capturing temporal dependencies, making this combination suitable for sequential image data. The architecture was built using 5 convolutional layers, an LSTM layer and the output layer as the dense layer with the activation function being sigmoid for the output layer as it's a classification problem. It was trained on 50 epochs, consisting of a batch size of 200 with 'Adam' optimizer.

#### **Artificial Neural Network (ANN):**

ANNs possess the versatility to learn intricate relationships between inputs and outputs through interconnected layers of neurons. The architecture involved 6 layers with 1 Flattening layer and 5 Dense layers. It was trained on 50 epochs, consisting of a batch size of 200 and activation function being sigmoid for the output layer and ReLu for the remaining layers.

**Decision Tree:** Decision Trees are machine learning methods that create a tree-like model of decisions by recursively partitioning the images into subsets based on the most important feature at each node. They select attributes that best separate the images into classes or minimize impurity measures such as Gini impurity or entropy. This process continues until a stopping criterion is met, creating a simple and interpretable model.

**Random Forest:** It's an ensemble learning method that involves the creation of multiple decision trees during the training. It works by constructing a multitude of decision trees and outputs the most frequent class label among the individual trees.

**Support Vector Machine (SVM):** SVMs use feature space to classify data by locating the hyperplane that efficiently divides several classes.

**YOLO:** YOLO is an object-detection method that uses class probabilities and bounding box forecasts for each grid cell to identify objects in pictures.

#### 3. Training

The data was split into a training set and a testing set during the training phase.

The machine learning models were trained using the training set. There were 5000 photos total, of which 2500 featured accidents and the remaining 2500 did not.

Each algorithm underwent iterative training on the training set, where it learned to discern patterns and relationships between the input features (i.e., image pixels) and corresponding labels (i.e., accident or non-accident).

For instance, in CNN with LSTM, the model was trained to extract spatial features using CNN layers and capture temporal dependencies using LSTM layers. In order to reduce the error between the anticipated and real labels, optimization techniques including gradient descent and backpropagation were used to modify the model's weights.

Similar training processes were tailored for other algorithms such as ANN, Random Forest, Decision Tree, SVM, and YOLO, optimizing their parameters to enhance their accuracy in classifying traffic accident images.

#### 4. Testing

Following the training phase, the trained models underwent evaluation using the testing set, which served as a separate dataset unseen during training. This step was vital to evaluate the models' generalization ability and estimate their performance on new, unseen data.

The testing set was given to each trained model, generating predictions based on learned patterns. The forecasted labels were then compared with the actual labels to calculate evaluation parameters such F1-score, recall, accuracy, and precision, providing quantitative measures of the models' classification performance.

#### 5. Interpretation using SHAP

Post-training and testing, the study proceeded to obtain the results from the trained machine learning models. The SHAP (SHapley Additive exPlanations) framework was employed for this purpose.

SHAP, an advanced tool in explainable AI, attributes importance to features, elucidating their contributions to the models' predictions. In this research, SHAP facilitated a deeper understanding

of how each feature (e.g., image characteristics) influenced the models' classification decisions regarding traffic accidents.

For instance, SHAP analysis could elucidate which spatial features obtained by CNN layers and temporal dependencies captured by LSTM layers contributed most to accident/non-accident classification. Similarly, for ANN, SHAP provided insights into the importance of different features in decision-making.

### IV. RESULTS

Presented below are the parameters of the implemented models.

Table 1: CNN + LSTM

| CNN + LSTM | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|------------|----------|-----------|--------------|----------|--------------|----------|--------------|
|            |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test       | 93.00%   | 93%       | 93%          | 93%      | 93%          | 93%      | 93%          |

Table 2: ANN

| ANN  | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|------|----------|-----------|--------------|----------|--------------|----------|--------------|
|      |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test | 92.90%   | 91%       | 91%          | 91%      | 91%          | 91%      | 91%          |

Table 3: CNN

| CNN  | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|------|----------|-----------|--------------|----------|--------------|----------|--------------|
|      |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test | 87.80%   | 82%       | 96%          | 96%      | 79%          | 89%      | 87%          |

Table 4: Decision Tree

| Decision Tree | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|---------------|----------|-----------|--------------|----------|--------------|----------|--------------|
|               |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test          | 83.33%   | 84%       | 83%          | 83%      | 84%          | 83%      | 83%          |

Table 5: SVM

| SVM  | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|------|----------|-----------|--------------|----------|--------------|----------|--------------|
|      |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test | 90%      | 89%       | 91%          | 91%      | 89%          | 90%      | 90%          |

Table 6: Random Forest

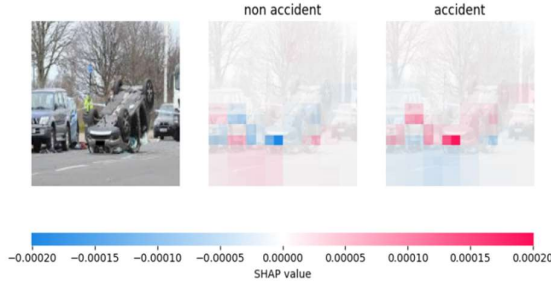
| Random Forest | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|---------------|----------|-----------|--------------|----------|--------------|----------|--------------|
|               |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test          | 89%      | 92%       | 86%          | 85%      | 92%          | 88%      | 89%          |

Table 7: YOLO v8

| YOLO v8 | Accuracy | Precision |              | Recall   |              | F1 Score |              |
|---------|----------|-----------|--------------|----------|--------------|----------|--------------|
|         |          | Accident  | Non-Accident | Accident | Non-Accident | Accident | Non-Accident |
| Test    | 91%      | 91%       | 90%          | 91%      | 90%          | 91%      | 90%          |

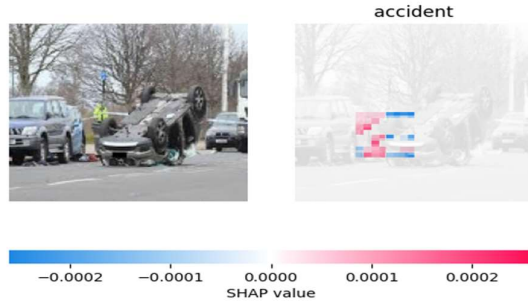
## V. MODEL INTERPRETATION USING SHAP

### CNN + LSTM



*Fig 1: Application of SHAP to interpret the CNN + LSTM model's decision-making process.*

### ANN



*Fig 2: Application of SHAP to interpret the ANN model's decision-making process.*

The interpretive ability of SHAP demonstrates the key role of diverse features in dictating the models'

predictions, particularly within the context of accident detection. These insights offer clarity into how the models distinguish between accident and non-accident imagery, thus strengthening their reliability and trustworthiness in critical scenarios. By pinpointing the specific features that drive the process of classification, SHAP empowers stakeholders to observe and validate the models' decision-making mechanisms, ultimately enhancing their efficacy in accident-related tasks.

## VI. DISCUSSIONS

Upon analysis, the CNN + LSTM architecture achieved an impressive accuracy of 93% on the testing set. High and positive SHAP values were observed in the region where accidents were detected, indicating the significance of these images' features when classifying the image as an accident. Conversely, negative SHAP values were observed in regions contributing to the image being classified as non-accident.

Similarly, the ANN architecture demonstrated promising results with a level of accuracy of 92.90% on the testing set, closely trailing the capabilities of the CNN + LSTM architecture. However, the SHAP values achieved from the ANN model were not as precise as those from the CNN + LSTM model. The highlighted regions seemed more evenly distributed between accident and non-accident classifications, with some regions responsible for accidents showing negative SHAP values, suggesting their effect on the image being classified as non-accident.

For optimal performance, the model should exhibit higher positive SHAP values to achieve a higher True Positive rate. Due to this, CNN + LSTM had a high precision score in comparison to ANN.

## VII. FUTURE WORK

For more advanced research avenues, enhancing model interpretability and addressing potential bias in the data is essential. Exploring the generalizability of the models across various geographical locations and environmental conditions could strengthen the robustness of accident detection systems. Additionally, integrating real-time data streams and contextual information guarantees enhancing the accuracy and reliability of these systems in real-world applications.

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